

1. First, you should look on the web, and if it can be done feasibly, look in stores to find possible competitors for the product you selected.

- Who is your competition?

Our competition is against regular, rubber pasta strainers that can attach onto a pot.

- What are the aspects of your product that are better/worse in terms of:

- features – Our product will be better in terms of features. It will include more features that will regulate the amount of steam coming out of the pot, and it will also include a handle cover so that users do not burn themselves. In addition, it will fold for easier storage.

- specifications – Our product will have more specifications, as there are more design components that need to be considered.

- target users – We will have a slightly smaller number of target users, as we are mostly trying to target users who want something better than a regular pasta strainer.

- costs – Our product will likely be more expensive simply because more material will be used, and the manufacturing process will become slightly more complicated.

2. Find on the web, any available Ergonomics based Task Analysis software then Tool/Product Assessment Questionnaire. You will have to import one of your product pictures. The survey can be broad; so in answering the questions, consider your product as one in which an individual uses and it is primary to their or other important activity. Upon completion of the survey, print your results. Examples of questions: Top 10 Product Survey Questions, with Examples [Typeform)

Results of survey:

Question	Response
How often do you use our products?	I use this product every time I cook pasta.
Which features are most valuable?	I like how it attaches to a pot. This is a lot more convenient than having to put the strainer in the sink.
How would you compare our products to our competitors' products?	I like this product better. I like the handle so that I do not burn my hands, and it is nice that it blocks the steam from hitting me.
What important features are we missing?	I think this product needs something to store it in.
What're you trying to solve by using our products?	By using this product, I am eliminating the possibility of burning myself with another strainer/pot.
Who else could find our products useful?	I think chefs could find this product useful.
How easy is it to use our product?	It is very easy to use this product, but sometimes it is difficult to get on the pot.
How would you rate the value for the money?	I think this is a good product for the price.

How likely are you to recommend our products to others?	I would recommend this product to others.
How could we improve our product to better meet your needs?	I think this product just needs something to store it in.

3. Next, you will discuss as a group the potential problems you have now acquired via the task analysis, market analysis, and tool assessment questionnaire. This discussion should help you to concentrate your approach for redesign and how you will review the literature for your improvements.

- **Market analysis:**

Competitors	[1] Kajaoel	[2] Emptycolor	[3] TJ POP	[4] Handy Housewares
Picture				
Features	Foldable silicone clip-on design, hands-free use, adjustable to different pot sizes	Food-grade silicone, clips to various cookware, heat resistant up to 450°F	Stainless steel, fits 10" pots, durable build	Simple plastic clip-on design, standard fit
Strengths	Space-saving, flexible silicone material, easy to use	Affordable, universal fit, straightforward use	Long lifespan, strong material	Affordable, easy to use (one-piece), simple structure
Weaknesses	No steam regulation and proper handle insulation	No folding mechanism, limited safety features	Heavy, conducts heat, no steam protection, not foldable	Minimal functionality, poor durability, no heat protection
Target Users	General home cooks seeking compact kitchen tools	Students or small households looking for inexpensive, simple tools	Experienced home cooks preferring sturdy metal tools	Casual or occasional cooks seeking low-cost tools
Price	\$8.99	\$7.75	\$6.99	\$8.99
Comparison with our Product	Advantages: Our design adds steam-control vents and heat-resistant handles. Disadvantage: Slightly higher production costs due to extra components.	Advantages: Our product adds folding storage and steam protection. Disadvantage: Their product is simpler and cheaper to produce.	Advantages: Ours is lighter, safer to handle, and adaptable. Disadvantage: Stainless steel versions have greater brand trust and longevity.	Advantages: Our design integrates safety, durability, and ergonomics. Disadvantage: Slightly higher price and design complexity.

Discussion:

Resources for Memo:

<https://journals.sagepub.com/doi/pdf/10.1177/0018720814553471>

<https://www.sciencedirect.com/science/article/pii/S0003687013000975>

<https://ebookcentral.proquest.com/lib/pitt-ebooks/reader.action?docID=726854&c=RVBVQg&ppg=10>

To: Instructor

From: Alexa, Aleia, Arth, Irvin, Amaris

Date: 22 Oct 2025

Subject: Market Analysis & Literature-Backed Ergonomic Review of the Clip-on Pot Strainer

After analyzing the current market, our group found that our clip-on pot strainer competes well with the products already out there. Similar items like Kajaoel and Handy Housewares are also silicone strainers that attach to different pot sizes, while brands like TJ POP use stainless steel for durability but sacrifice weight and heat protection [1, 3, 4]. What makes our design different is that it focuses on user comfort and safety. We added an insulated handle, better steam control, and a lightweight design to make the tool easier and safer to use. Even though there are several competitors, the market for clip-on strainers is small and scattered, so it's hard to find exact data on market share. Most products are inexpensive and simple to make, ranging from \$6.99–\$8.99, and so we believe we can add small ergonomic improvements without raising the cost or changing who it's made for [1, 2, 3, 4]. Based on the research, an ideal handle diameter is between 30 and 35 millimeters, and having a textured, slightly soft surface helps reduce how much force users need to grip the tool. This means we can easily make small material or shape adjustments that improve comfort without expensive redesigns.

According to Szalma (2014), human–technology interaction isn't only about physical comfort but also about maintaining user motivation and reducing cognitive strain. A design that supports users' autonomy, competence, and sense of connection can increase engagement and safety. In this sense, our strainer's simple, intuitive use promotes autonomy by letting users control how and where to attach it, while the lightweight build supports competence by matching users' physical abilities and minimizing fatigue.

While comparing products, we noticed a few clear human factor issues that other brands haven't solved. Steam burns are one of the most common kitchen injuries, which makes it important to control where steam escapes. Some competitors also use metal clips that get hot to the touch, increasing the chance of contact burns. The heavier stainless designs require more wrist strength and can force awkward hand positions while pouring. Others have small, slippery handles that are harder to hold, especially when wet. These aren't just marketing problems, they are real ergonomic concerns that deal with posture, force, and safety.

Incorporating Szalma's (2014) motivational design principles helps address these ergonomic and psychological concerns together. For example, by improving control over steam direction and reducing grip strain, we are also supporting the user's sense of competence and autonomy. When tools create a feeling of safety and predictability, users become more confident and motivated to use them correctly, reducing frustration and potential errors.

Our tool is body-friendly because it's light, easy to set up, and keeps the wrist and arm in a comfortable position while using it. Based on the product design principles we discussed in class, our design fits the user rather than forcing the user to adapt to it. It reduces effort by keeping the tool light and simple to grip. It also helps users keep a straight wrist, which lowers strain on the forearm. The insulated handle and steam vents protect against burns, and the simple setup makes the tool intuitive to use, reducing frustration and error. These choices all support user comfort and safety, which are the main goals of good product design.

In line with Szalma's findings, designs that reduce user frustration and improve intuitive understanding also enhance intrinsic motivation, encouraging consistent, confident use rather than short-term or error-prone behavior. Our product, therefore, promotes both ergonomic comfort and motivational engagement, fostering an overall positive user experience.

The strainer also works well for a wide range of users. It clips easily onto different pot sizes, so people don't have to use much strength or awkward movements. Users with smaller hands or less grip strength might find it a little harder to hold if the handle is too large, but that can be fixed by making the grip slightly slimmer or more oval-shaped. The literature suggests designing handles that fit between the 5th and 95th percentile of hand sizes, so we plan to adjust the grip so it's comfortable for almost everyone.

From a motivational design perspective, this inclusivity also promotes relatedness, one of Szalma's (2014) key components, by ensuring the product is accessible to diverse users and does not exclude those with limited dexterity or strength. Creating tools that fit a broader range of people helps users feel more connected and supported by the design itself.

When using the strainer, the main body parts involved are the hands, fingers, wrists, and forearms. These areas handle the gripping and pouring motion. There's also a chance of steam hitting the back of the hand or arm while draining hot water, which supports why our design focuses on steam control and insulation.

From our results, the tool should be redesigned with a few main improvements in mind. The grip could be made more comfortable and balanced, with a textured coating that prevents slipping. We can add better steam vents that direct heat away from the hand and a small shield above the handle to stop splashes. Adjusting the angle of the handle slightly will help users keep a neutral wrist position while pouring. Literature also shows that lighter tools reduce fatigue and improve control, so keeping the weight under half a kilogram will make a big difference. Adding small tactile features like a click when the clip locks or the vent adjusts would make the tool feel more intuitive to use. These changes all match human factor recommendations and would not require big manufacturing costs.

Integrating Szalma's motivational framework into these updates further ensures that the product not only performs efficiently but also encourages users to feel capable, autonomous, and safe. By giving tactile feedback (like a click) and predictable responses to user input, the strainer increases the user's sense of competence and trust in the tool, key psychological factors that improve overall satisfaction and usability.

Overall, our analysis and research show that simple ergonomic improvements can make a big difference in user comfort and safety. By focusing on handle design, steam direction, and posture, our product can outperform others on the market while staying affordable. Furthermore, aligning our design with motivational principles outlined by Szalma (2014) supporting user autonomy, competence, and relatedness, ensures that the product not only meets ergonomic standards but also sustains user engagement and enjoyment. These updates will help make our redesigned strainer safer, easier to handle, and more enjoyable to use for everyday cooking.